



GENERAL GUIDELINES

AND LESSONS LEARNT FOR FUTURE CASE STUDIES



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Creation process

The process of creating a new case study begins with a **handwritten draft**. It includes a vague idea of what wants to be communicated in the animation: notes, equations, graphs and illustrations. This draft is then digitalised to a more formal document in LaTeX format, using an online LaTeX editing software called Overleaf. This **digitalised draft** contains the exact equations that appear in the video, and follow the same storyline.

Next it is time to use the digitalised draft to create the animation itself. The video is usually composed of two different types of animations. The first type – **mathematical animation** – is developed using Manim: a community-maintained Python library for creating mathematical animations. This means that all animations of this type are written as code, and later rendered. This software is used for the parts of the video containing equations, derivations, graphs, and everything connected to mathematics. The other type – **aircraft animation** – is developed using Blender: an open-source software for 3D modelling and animation. This type of animation is created via graphical user interface and contains the visuals of the aircraft Cessna Citation II for the purpose of showing real-life simulation of a concept in the video.

Given the animations and the draft, a **voiceover script** can be written. It is essentially the text that accompanies the animation. It is written based on the draft, however it is also adapted to suit the animation well. The voiceover audio is recorded in a professional studio by one of our lecturers.

The two types of animations and the voiceover audio are composed into a deliverable **Case Study video** in a video editing software DaVinci Resolve. There the animations are cut, assembled together and matched with the audio. Such a deliverable is then reviewed, and a lot of feedback is gathered. The development process proceeds in an iterative fashion, adapting the draft, animations, and the voiceover accordingly, until the final version is created.

Software used

The animations were primarily created using the python Manim library, and blender. Both of the tools are open source which aligns with the idea of open educational materials. Manim was mainly used for visuals containing mathematical equations and geometrical representations. Blender was used only for 3D scenes involving more complex objects. Both of the tools were sufficient to complete the case studies in a satisfactory manner, however it is worth noting that they are not flawless. Manim, being a very niche and community-driven tool, still contains a lot of bugs, however they only appear in more complex tasks, such as 3D rendering. Manim was chosen due to its programmistic approach, as all of the animations are written in code. It is therefore possible to be very exact in showing the results of mathematical operations. Nevertheless, at many points during the project some hesitation emerged whether the case studies could benefit from more professional animation software, such as Unity.

Visual semantics

Visual elements should be used consistently and intentionally throughout a case study. Every object, movement, transformation, and graphical effect should have a reason to exist and contribute to the explanation.

Objects that represent the same concept should retain the same appearance across different scenes, and better case studies. When an object is introduced using a particular colour or visual style, this representation should remain consistent throughout the video, and the series. Changes in size, position, colour, or opacity should indicate a meaningful change in the underlying concept. For example, colour for highlighting, colour coding, or grouping, and opacity for indicating importance or priority.

Animations should be used to enhance the explanation. For example:

- variables can be transformed into expanded expressions, to signify equivalence.
- terms can disappear signifying their equal to 0 in case of addition, or 1 in terms of multiplication. Consequently same for matrices and vectors.
- equations can be written left to right to simulate traditional learning environment, rather than appear all at once.
- vector images (SVG) can be transformed into simpler mathematical shapes, e.g. when modelling an airplane as a mass-spring system.
- matrix or vector equations can be visually expanded into a system of equations.
- Manim's `wiggle` or `Indicate` animations can be used to highlight the concept that is currently being mentioned in the animation.
- Manim's `ApplyMatrix` can be used to show the transformation in an animated way, rather than just showing the before and after.
- a camera movement to the side can be used to start a new discussion on the side, auxiliary to what was being covered before.
- and many, many more...

The general rule of thumb is to introduce as few new objects from scratch as possible. The previously introduced objects should evolve into next ones by animational transformations. Maintaining visual continuity reduces the cognitive effort required to follow a derivation.

Colour coding

Colour coding should be established during the early stages of production and applied consistently across the entire case study. Colours may be assigned to variables, vectors, or other recurring concepts. The same colour should not be used for unrelated elements unless the connection is intentional.

A limited colour palette is preferable to a large number of colours. Colour should not be the only method used to communicate meaning. Differences should also be indicated through labels, line styles, symbols, shapes, or positioning. This is particularly important for viewers with colour-vision deficiencies.

The colours used were selected from the TU Delft house style colour guide and should provide sufficient contrast with the background and remain readable when the video is displayed on different screens or projected in a lecture room. The appearance of equations, graphs, and annotations should therefore be tested at the intended output resolution before the final rendering stage.

Matching visuals with the voiceover

The visuals and voiceover should be complementary. The animation should support what is being said without simply reproducing every word as text on the screen. Important equations, definitions, assumptions, and conclusions may be displayed, while the voiceover provides context and explains their meaning.

New visual information should normally appear at, or shortly before, the moment it is introduced in the voiceover. Introducing it too early may distract the viewer, while introducing it too late can make the explanation difficult to follow. Visual elements should also remain on the screen long enough for the viewer to read and understand them before the animation moves to the next step.

Ideally, the voiceover script and animation should be developed together. Writing the complete script only after the animation has been finalised may lead to timing problems, while recording the voiceover before the visuals have been planned may restrict the animation unnecessarily. A preliminary script or temporary voice recording can therefore be used during production, followed by the final studio recording once the structure and timing are sufficiently stable.

During editing, the animation should be synchronised with the final audio at the level of individual sentences and visual actions.

Mathematical notation and accuracy

This point is especially important because the videos teach technical material. Mathematical notation should be consistent with the one used in lectures, GraspLe exercises, and the LA book. Moreover, for the ease of production it should remain consistent between the handwritten draft, LaTeX document, Manim scenes, voiceover script, and final video. Variables should not change symbols, colours, or definitions without explanation. Equations should be checked independently before animation begins, as correcting an error late in the process may require expensive changes to the script, rendered scenes, and audio. Units, coordinate systems, sign conventions, and assumptions should be established before creating a case study series.

Consistent visual identity

All case studies should share one visual identity. A common template should be used for all case studies, defining the background, typography, colour palette, equation formatting, graph style, line thickness, and transition conventions. Reusing these standards improves consistency across videos and reduces the amount of design work required for each new case study. Templates should nevertheless remain flexible enough to accommodate the specific requirements of particular topics.

Accessibility

Accessibility should be considered throughout the production process. Text and equations are sufficiently large, and strong contrast was maintained between foreground and background elements. Colour is not the sole means of distinguishing information, the videos can be watched and fully understood in grayscale. The voiceover is supported by subtitles.

Technical workflow and file organisation

Manim, Blender, audio, and DaVinci Resolve produce many files and versions. A standard folder structure and naming convention should be used for scripts, Blender projects, audio recordings, rendered footage, reference material, and final exports. Source files should be separated from generated files, and scene names should correspond across the storyboard, code, renders, and editing timeline. Version control should be used for Manim code and LaTeX files where possible. Major rendered or edited versions should also be labelled clearly to prevent outdated material from being used.

Final lessons learnt and recommendations

The development of the educational case studies requires close coordination between subject knowledge, visual design, narration, and technical production. The most effective scenes were generally those that were planned around a single learning objective and used animation to reveal a relationship or process that would be difficult to communicate using static material alone. Future projects should devote more time to storyboarding, and establish visual and technical conventions before production begins. Mathematical content should be approved before final rendering and voiceover recording. Reusable templates and code components should also be developed to reduce duplication and improve consistency across case studies. The choice of software should be based on the educational and technical requirements of each scene. Manim remains particularly suitable for precise mathematical animation, while Blender is more appropriate for complex three-dimensional objects and physical demonstrations. Introducing another platform, such as Unity, should only be considered when it provides a specific capability that cannot be achieved efficiently using the existing workflow, as adopting additional software would also increase the complexity of production.